G:/code/vhdl/mini_cpu.vhd (/tb_mini_cpu/uut) - Default

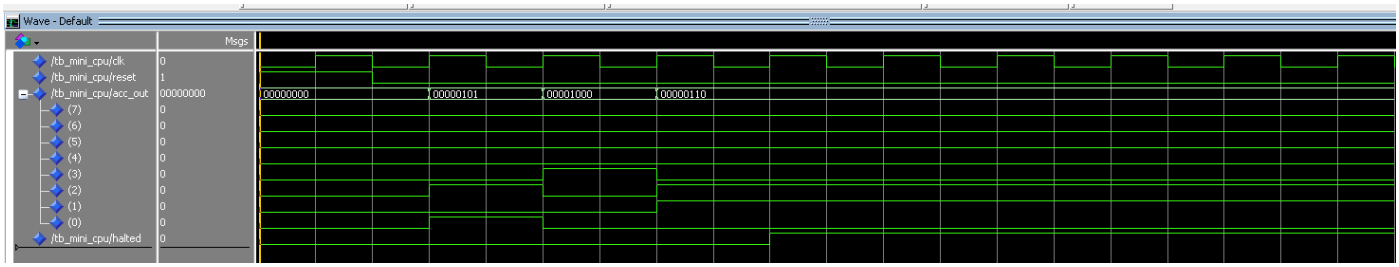
```
Ln#
 1  library ieee;
 2  use ieee.std_logic_1164.all;
 3  use ieee.numeric_std.all;
 4
 5  entity mini_cpu is
 6      port (
 7          clk, reset : in std_logic;
 8          acc_out : out std_logic_vector(7 downto 0);
 9          halted : out std_logic
10      );
11  end mini_cpu;
12
13  architecture behavioral of mini_cpu is
14
15      signal acc : unsigned(7 downto 0) := (others => '0');
16      signal pc : integer range 0 to 3 := 0;
17      signal stop : std_logic := '0';
18
19  begin
20
21      process(clk, reset)
22      begin
23          if reset = '1' then
24              acc <= (others => '0');
25              pc <= 0;
26              stop <= '0';
27
28          elsif rising_edge(clk) then
29              if stop = '0' then
30                  case pc is
31                      when 0 =>
32                          acc <= to_unsigned(5, 8);
33
34                      when 1 =>
35                          acc <= acc + 3;
36
37                      when 2 =>
38                          acc <= acc - 2;
39
40                      when 3 =>
41                          stop <= '1';
42                  end case;
43
44                  if pc < 3 then
45                      pc <= pc + 1;
46                  end if;
47              end if;
48          end if;
49      end process;
50
51      acc_out <= std_logic_vector(acc);
52      halted <= stop;
53
54  end behavioral;
```

Program 1: VHDL Code for Simple Traffic Light Controller

G:/code/vhdl/tb_mini_cpu.vhd (/tb_mini_cpu) - Default

```
Ln#
 1  library ieee;
 2  use ieee.std_logic_1164.all;
 3
 4  entity tb_mini_cpu is
 5  end tb_mini_cpu;
 6
 7  architecture test of tb_mini_cpu is
 8
 9      signal clk : std_logic := '0';
10      signal reset : std_logic := '0';
11      signal acc_out : std_logic_vector(7 downto 0);
12      signal halted : std_logic;
13
14  begin
15
16      uut: entity work.mini_cpu
17          port map (
18              clk => clk,
19              reset => reset,
20              acc_out => acc_out,
21              halted => halted
22          );
23
24      clk <= not clk after 5 ns;
25
26      process
27      begin
28          reset <= '1';
29          wait for 10 ns;
30
31          reset <= '0';
32          wait for 50 ns;
33
34          wait;
35      end process;
36
37  end test;
```

Program 2: VHDL Testbench for Simple Traffic Light Controller



Testbench Output: Simulation Waveform of Simple Traffic Light Controller